

CEE+

Causal Explanation Engine

Project overview for intern · U.S. Army Research Laboratory · Lead: Sunny Anjon

WHAT IS CEE+

AI models can look at an image and write confident reports about what is happening, what the dangers are, and what to do about it. CEE+ is a research project that checks whether those reports are actually grounded in the AI's reasoning, or whether they only sound right on the surface.

We work with disaster scenes (mostly fires) and use the AI's recommendations as the test case. If the AI wrote a recommendation because of a fire, then removing the fire from the scene should change the recommendation. If the recommendation does not change, the AI was not really reasoning about the fire.

WHY THIS MATTERS

AI tools are being used to support real decisions in emergency response, military operations, and other settings where being wrong has serious consequences. The output of a modern AI can look completely trustworthy while being only loosely connected to the actual scene. Operators acting on those reports can be misled in ways that are hard to catch.

CEE+ gives an honest way to check whether a particular AI report is grounded in real reasoning or only in plausible-sounding language. That tells the operator when to trust the report and when to take a second look before acting.

WHAT ALREADY EXISTS

A working pipeline that takes an image and a short caption, runs them through the AI, and produces:

- a structured analysis of the scene (objects, threats, recommendations, reasoning)
- a trust score with Low / Moderate / High bands
- flags for known AI failure patterns (model agreeing too quickly, hedging real threats, etc.)
- a batch report that summarizes results across many scenes

All of the above is the baseline pass. It tells us what the AI thinks. It does not tell us whether the AI was actually reasoning about the scene. That is the part you will build.

WHAT YOU WILL WORK ON

You will build the intervention step. The idea is to take a scene, remove one hazard (either by editing the caption to remove the mention, or by editing the image to remove the visual evidence, or both), and run the AI again on the modified scene. We then compare the two outputs to see what changed.

If the AI's reasoning was real, removing the hazard should change the report in predictable ways. If the report stays the same, the original reasoning was not grounded. This is the core experiment of the project.